



DEPARTMENT OF COMPUTING
AND CONTROL ENGINEERING

ДЕПАРТМАН ЗА РАЧУНАРСТВО
И АУТОМАТИКУ

ACSE

Odsek za automatiku, geomatiku
i upravljanje sistemima

Sub- department for Automatic Control
and System Engineering

Linearani matematički modeli

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

Osobine linearnih modela

Osobine linearnih modela

Model je linearan ako ispunjava osobine:

1. superpozicija

$$y_1 + y_2 = f(u_1 + u_2) = f(u_1) + f(u_2)$$

$$y_1 = f(u_1), \quad y_2 = f(u_2)$$

2. homogenost

$$m \cdot y = f(m \cdot u) = m \cdot f(u)$$

$$y = f(u)$$

3. stacionarnost

$$y(t - \tau) = f(u(t - \tau))$$

$$y = f(u)$$

Primer 1

- Da li je model linearan ?

$$y(t) = a \cdot u(t)$$

- Superpozicija:

$$u = u_1 + u_2$$

$$y = a(u_1 + u_2) = au_1 + au_2$$

$$y = y_1 + y_2$$

$$y_1 = au_1, \quad y_2 = au_2$$

- Homogenost:

$$a(mu) = m(au) = my$$

- Stacionarnost:

$$y(t - \tau) = au(t - \tau)$$

- Jeste linearan.

Primer 2

- Da li je linearan?

$$y = au + b$$

- Nije! Ne važi princip homogenosti.

$$m(y) = m(au + b) = m(au) + m(b) \neq a(mu) + b$$

Primer 3

- Da li je linearan?

$$y = au^2$$

- Nije! Ne važi princip superpozicije.

$$a(u_1 + u_2)^2 = au_1^2 + au_2^2 + 2au_1u_2 \neq au_1^2 + au_2^2$$

Primer 4

- Da li je model linearan ?

$$\ddot{y}(t) + a\dot{y}(t) + by(t) = cu(t)$$

- Superpozicija:

$$\ddot{y}_1(t) + a\dot{y}_1(t) + by_1(t) = cu_1(t)$$

$$\ddot{y}_2(t) + a\dot{y}_2(t) + by_2(t) = cu_2(t)$$

$$u = u_1 + u_2$$

$$y = y_1 + y_2$$

$$\frac{d^2}{dt^2} (y_1 + y_2) + a \frac{d}{dt} (y_1 + y_2) + b(y_1 + y_2) = c(u_1 + u_2)$$

- Homogenost:

$$\frac{d^2}{dt^2} (my) + a \frac{d}{dt} (my) + b(my) = c(mu)$$

- Stacionarnost:

$$\ddot{y}(t - \tau) + a\dot{y}(t - \tau) + by(t - \tau) = cu(t - \tau)$$

- Jeste linearan.

Primer 5

- Da li je model linearan ?

$$\ddot{y} + \mu(1 - y^2)\dot{y} + y = u$$

- Superpozicija:

$$\ddot{y}_1 + \mu(1 - y_1^2)\dot{y}_1 + y_1 = u_1$$

$$\ddot{y}_2 + \mu(1 - y_2^2)\dot{y}_2 + y_2 = u_2$$

$$u = u_1 + u_2$$

$$y = y_1 + y_2$$

$$\frac{d^2}{dt^2} (y_1 + y_2) + \mu(1 - (y_1 + y_2)^2) \frac{d}{dt} (y_1 + y_2) + (y_1 + y_2) =$$

$$(\ddot{y}_1 + \ddot{y}_2) + \mu(1 - (y_1^2 + 2y_1y_2 + y_2^2)) (\dot{y}_1 + \dot{y}_2) + (y_1 + y_2) =$$

$$(\ddot{y}_1 + \ddot{y}_2) + \mu(\dot{y}_1 + \dot{y}_2) - \mu(y_1^2 + 2y_1y_2 + y_2^2)(\dot{y}_1 + \dot{y}_2) + (y_1 + y_2) =$$

$$\ddot{y}_1 + \mu\dot{y}_1 - \mu y_1^2 \dot{y}_1 + y_1 + \ddot{y}_2 + \mu\dot{y}_2 - \mu y_2^2 \dot{y}_2 + y_2 - \mu(2y_1y_2\dot{y}_1 + y_2^2\dot{y}_1 + \dots) = u_1 + u_2$$

Kada se skрати ostaje „višak“

$$\mu(2y_1y_2\dot{y}_1 + y_2^2\dot{y}_1 + \dots) \neq 0$$

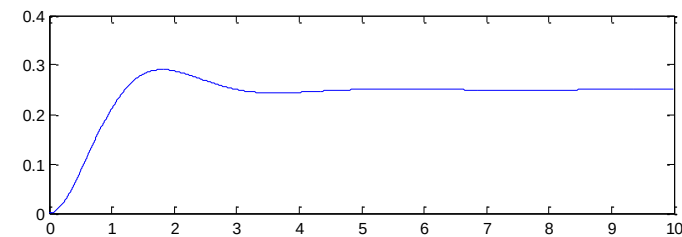
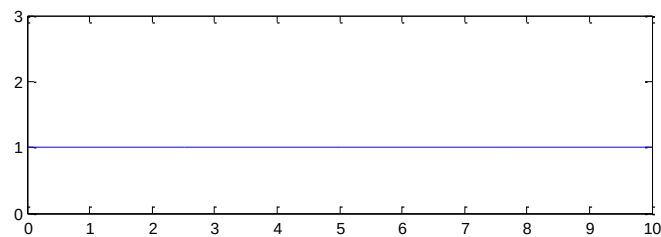
- Nije linearan jer princip superpozicije nije zadovoljen.

Linearan model: primeri odziva (desno) na pobudu (levo)

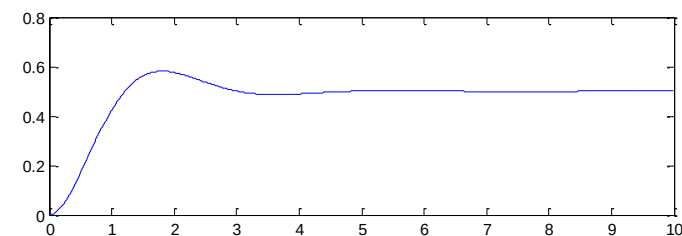
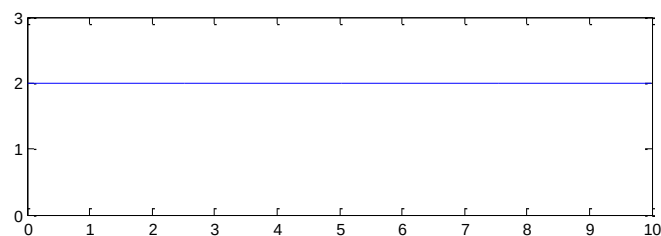
Ulazi

Izlazi

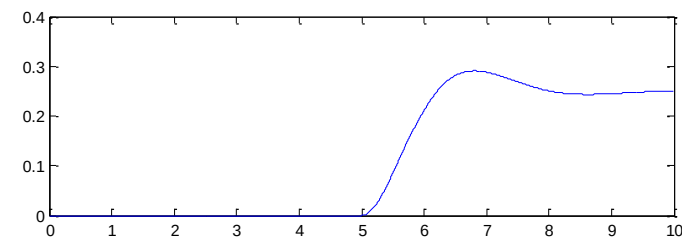
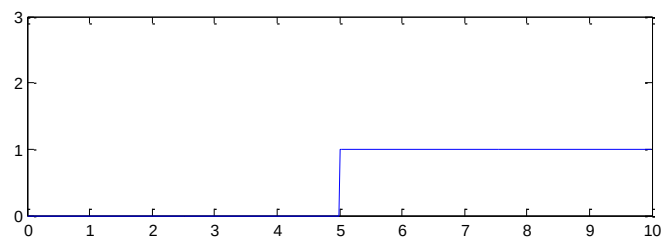
a)



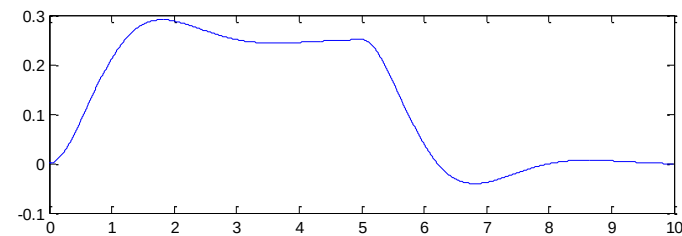
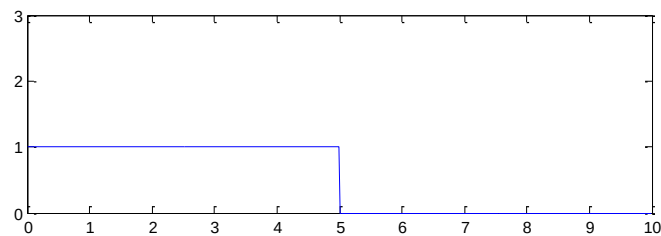
b)



c)



d)



Kakav model posmatramo?

- Matematički model:
 - dinamički
 - vremenski kontinualan
 - linearan
 - sa koncentrisanim parametrima
 - deterministički
- Opisuje se običnim linearnim diferencijalnim jednačinama
 - Sistem diferencijalnih jednačina 1. reda
- Primena Laplasove transformacije



Linearan matematički model
u prostoru stanja

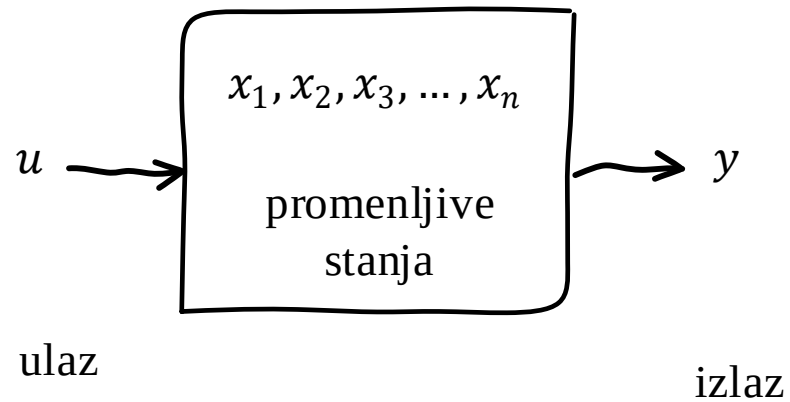


Funkcija prenosa

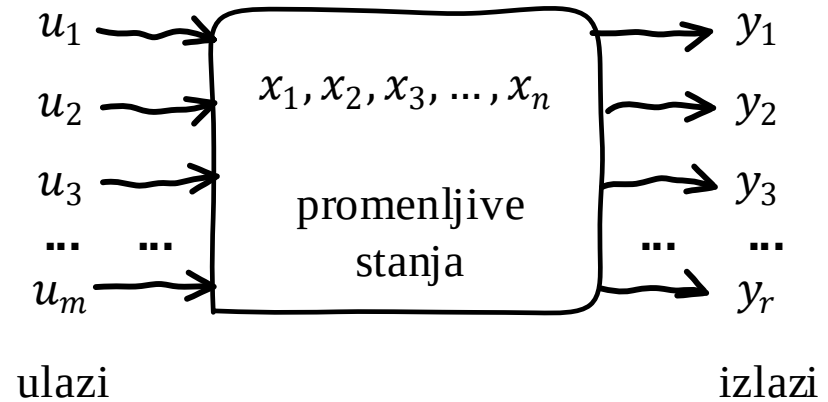
Matematički model u prostoru stanja

Promenljive za opis modela

Jedan ulaz – jedan izlaz



Više ulaza/izlaza – *multivarijabilni* model



Uvodimo vektore:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \dots \\ u_m \end{bmatrix} \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_r \end{bmatrix}$$

Linearan matematički model u prostoru stanja

u razvijenom obliku:

Izvodi (promene u vremenu) promenljivih stanja

$$\begin{aligned}\dot{\hat{x}}_1(t) &= a_{11}\hat{x}_1(t) + \dots + a_{1n}\hat{x}_n(t) + b_{11}\hat{u}_1(t) + \dots + b_{1m}\hat{u}_m(t) & \hat{x}_1(0) &= \hat{x}_{10} \\ \dot{\hat{x}}_2(t) &= a_{21}\hat{x}_1(t) + \dots + a_{2n}\hat{x}_n(t) + b_{21}\hat{u}_1(t) + \dots + b_{2m}\hat{u}_m(t) & \hat{x}_2(0) &= \hat{x}_{20} \\ &\dots & \dots & \\ \dot{\hat{x}}_n(t) &= a_{n1}\hat{x}_1(t) + \dots + a_{nn}\hat{x}_n(t) + b_{n1}\hat{u}_1(t) + \dots + b_{nm}\hat{u}_m(t) & \hat{x}_n(0) &= \hat{x}_{n0}\end{aligned}$$

Izlazi modela

$$\begin{aligned}\hat{y}_1(t) &= c_{11}\hat{x}_1(t) + \dots + c_{1n}\hat{x}_n(t) + d_{11}\hat{u}_1(t) + \dots + d_{1m}\hat{u}_m(t) \\ \hat{y}_2(t) &= c_{21}\hat{x}_1(t) + \dots + c_{2n}\hat{x}_n(t) + d_{21}\hat{u}_1(t) + \dots + d_{2m}\hat{u}_m(t) \\ &\dots \\ \hat{y}_r(t) &= c_{r1}\hat{x}_1(t) + \dots + c_{rn}\hat{x}_n(t) + d_{r1}\hat{u}_1(t) + \dots + d_{rm}\hat{u}_m(t)\end{aligned}$$

„kapice“ su bitne - imaju značenje promena (razlika) vrednosti u odnosu na nominalne vrednosti.

Linearan matematički model u prostoru stanja

u vektorskom zapisu

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A} \cdot \hat{\mathbf{x}}(t) + \mathbf{B} \cdot \hat{\mathbf{u}}(t), \quad \hat{\mathbf{x}}(0) = \mathbf{x}_0$$

$$\hat{\mathbf{y}}(t) = \mathbf{C} \cdot \hat{\mathbf{x}}(t) + \mathbf{D} \cdot \hat{\mathbf{u}}(t)$$

$$\hat{\mathbf{u}} = \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \dots \\ \hat{u}_m \end{bmatrix}, \quad \hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \dots \\ \hat{y}_r \end{bmatrix}, \quad \hat{\mathbf{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \dots \\ \hat{x}_n \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & \dots & & \dots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \dots & \dots & & \dots \\ c_{r1} & c_{r2} & \dots & c_{rn} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1m} \\ d_{21} & d_{22} & \dots & d_{2m} \\ \dots & \dots & & \dots \\ d_{r1} & d_{r2} & \dots & d_{rm} \end{bmatrix}$$

Primer iz mehaničkih sistema – ranije rađeno

(oznake su promenjene radi usaglašavanja. Dodatno, neka su izlazi pozicije iz primera.)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_1+k_2}{m_1} & -\frac{c}{m_1} & \frac{k_2}{m_1} & \frac{c}{m_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{m_2} & \frac{c}{m_2} & -\frac{k_2}{m_2} & -\frac{c}{m_2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_2} \end{bmatrix} u$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

(kapice nisu napisane)

(voditi brigu o osobinama matričnih operacija, tj. množenje matrica nije komutativno.)

(In)varijantan model

- Invarijantan model

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A} \cdot \hat{\mathbf{x}}(t) + \mathbf{B} \cdot \hat{\mathbf{u}}(t), \quad \hat{\mathbf{x}}(0) = \mathbf{x}_0$$

$$\hat{\mathbf{y}}(t) = \mathbf{C} \cdot \hat{\mathbf{x}}(t) + \mathbf{D} \cdot \hat{\mathbf{u}}(t)$$

- Model nije vremenski promenljiv jer su matrice $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ sa konstantnim koeficijentima.
- Primetiti da se vrednosti signala ulaza, izlaza i promenljivih stanja menjaju sa vremenom.

- Vremenski promenljiv model:

$$\mathbf{A} \equiv \mathbf{A}(t), \mathbf{B} \equiv \mathbf{B}(t), \mathbf{C} \equiv \mathbf{C}(t), \mathbf{D} \equiv \mathbf{D}(t)$$

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}(t) \cdot \hat{\mathbf{x}}(t) + \mathbf{B}(t) \cdot \hat{\mathbf{u}}(t), \quad \hat{\mathbf{x}}(0) = \mathbf{x}_0$$

$$\hat{\mathbf{y}}(t) = \mathbf{C}(t) \cdot \hat{\mathbf{x}}(t) + \mathbf{D}(t) \cdot \hat{\mathbf{u}}(t)$$

Transformacija dif. jedn. višeg reda u sistem dif. jednačina 1. reda

- Može se uraditi na mnogo načina, a ovde je prikazan jedan način ...
- Posmatra se dif. jedn. n -tog reda (linearna, obična):

$$y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1\dot{y} + a_0y = u(t)$$

- Uvode se smene:

$$x_1 = y$$

$$x_2 = \dot{y}$$

$$x_3 = \ddot{y}$$

...

$$x_n = y^{(n-1)}$$

- Tada je

$$\dot{x}_1 = \dot{y} = x_2$$

$$\dot{x}_2 = \ddot{y} = x_3$$

$$\dot{x}_3 = \ddot{\ddot{y}} = x_4$$

...

$$\dot{x}_n = y^{(n)} = -a_{n-1}x_{n-1} - \dots - a_1x_2 - a_0x_1 + u(t)$$

U literaturi postoji nekoliko uobičajenih načina ove transformacije i nazivaju se kanonske forme.

Pored njih, mogu se upotrebiti i druge smene, npr. (radi ideje) $x_1 = y$, $x_2 = \dot{y} + 1$, $x_3 = \dot{y} - \ddot{y}$...

Transformacija dif. jedn. višeg reda u sistem dif. jednačina 1. reda (2)

$$\dot{x}_1 = \dot{y} = x_2$$

$$\dot{x}_2 = \ddot{y} = x_3$$

$$\dot{x}_3 = \ddot{\ddot{y}} = x_4$$

...

$$\dot{x}_n = y^{(n)} = -a_{n-1}x_n - \dots - a_1x_2 - a_0x_1 + u(t)$$

- U matričnom obliku (oblik je „prepoznatljiv“):

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dots \\ 1 \end{bmatrix} \cdot u(t), \quad \mathbf{x}(0) = \begin{bmatrix} y(0) \\ \dot{y}(0) \\ \dots \\ y^{(n-1)}(0) \end{bmatrix}$$

$$y(t) = [1 \quad 0 \quad \dots \quad 0] \cdot \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} + [0] \cdot u(t)$$

Funkcija prenosa

Funkcija prenosa

- Obična linearna diferencijalna jednačina modela 1 ulaz – 1 izlaz

$$\begin{aligned} y^{(n)} + a_{n-1}y^{(n-1)}(t) + \dots + a_2\ddot{y}(t) + a_1\dot{y}(t) + a_0y(t) \\ = b_mu^{(m)}(t) + b_{m-1}u^{(m-1)}(t) + \dots + b_1\dot{u}(t) + b_0u(t), \quad n \geq m \end{aligned}$$

- Primeni se Laplasova transformacija

– model se nalazi u radnoj tački (sve početne vrednosti su nula!)

$$\begin{aligned} s^nY(s) + a_{n-1}s^{n-1}Y(s) + \dots + a_2s^2Y(s) + a_1sY(s) + a_0Y(s) \\ = b_ms^mU(s) + b_{m-1}s^{m-1}U(s) + \dots + b_1sU(s) + b_0U(s) \end{aligned}$$

- Dobije se funkcija prenosa

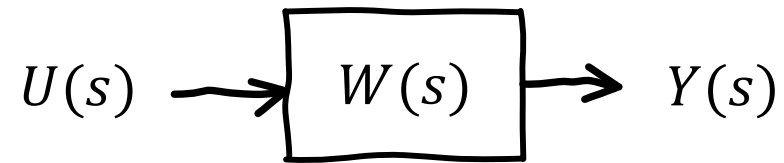
$$\frac{Y(s)}{U(s)} = W(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_2s^2 + a_1s + a_0}$$

Funkcija prenosa (2)

- Definicija:

Funkcija prenosa je odnos Laplasove transformacije izlaznog ($Y(s) = \mathcal{L}\{y(t)\}$) i ulaznog ($U(s) = \mathcal{L}\{u(t)\}$) signala, uz pretpostavku da su svi početni uslovi jednaki nuli i da je $u(t) = y(t) \equiv 0$ za svako $t < 0$.

- uzima samo odnos ulaz-izlaz
- ne pruža nikakvu informaciju o unutrašnjoj strukturi i ponašanju sistema



- Predstavlja se količnikom polinoma

$$W(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_2 s^2 + a_1 s + a_0} = \frac{P(s)}{Q(s)}$$

Primer

- Kako glasi funkcija prenosa amortizera?

$$m \frac{d^2 x(t)}{dt^2} + c \frac{dx(t)}{dt} + k \cdot x(t) = f(t)$$
$$x(0) \equiv 0, \quad \frac{dx}{dt}(0) \equiv 0$$

$$ms^2 X(s) + csX(s) + kX(s) = F(s)$$

$$W(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

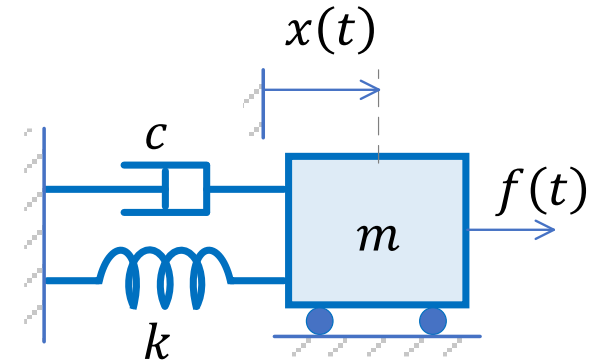
- Kako glasi odziv sistema ako je pobuda $f(t) = h(t) - h(t - 2)$?

$$m = 5, \quad c = 6, \quad k = 1$$

$$W(s) = \frac{1}{5s^2 + 6s + 1}, \quad F(s) = \frac{1}{s}(1 - e^{-2s})$$

$$X(s) = W(s)F(s) = \frac{1}{5} \frac{1 - e^{-2s}}{s(s+1)(s+0.2)} = \left(\frac{1}{s} + \frac{0.25}{s+1} - \frac{1.25}{s+0.2} \right) (1 - e^{-2s})$$

$$x(t) = (1 + 0.25e^{-t} - 1.25e^{-0.2t})h(t) - (1 + 0.25e^{-(t-2)} - 1.25e^{-0.2(t-2)})h(t-2)$$



Funkcija prenosa – faktorizovan oblik

$$W(s) = \frac{P(s)}{Q(s)} = k \frac{(s - r_1)(s - r_2) \dots (s - r_m)}{(s - t_1)(s - t_2) \dots (s - t_n)}$$

gde su:

- k – pojačanje ($k = b_m$)
- $r_1, r_2, \dots, r_m$ su **nule** sistema tj. koreni polinoma brojioca $P(s)$
- $t_1, t_2, \dots, t_n$ su **polovi** sistema $(s - t_1)(s - t_2) \dots (s - t_n)$
- Polovi sistema određuju karakter ponašanja sistema.
Dobijaju se rešavanjem jednačine

$$Q(s) = 0$$

Funkcija prenosa multivarijabilnog sistema

- Multivarijabilan sistem ima više ulaza i/ili izlaza
- Posmatraju se svi parovi ulaz-izlaz

$$G_{kj}(s) = \frac{Y_k(s)}{U_j(s)}, \quad k = 1, 2, \dots, r, \quad j = 1, 2, \dots, m.$$

- Nakon primene principa superpozicije

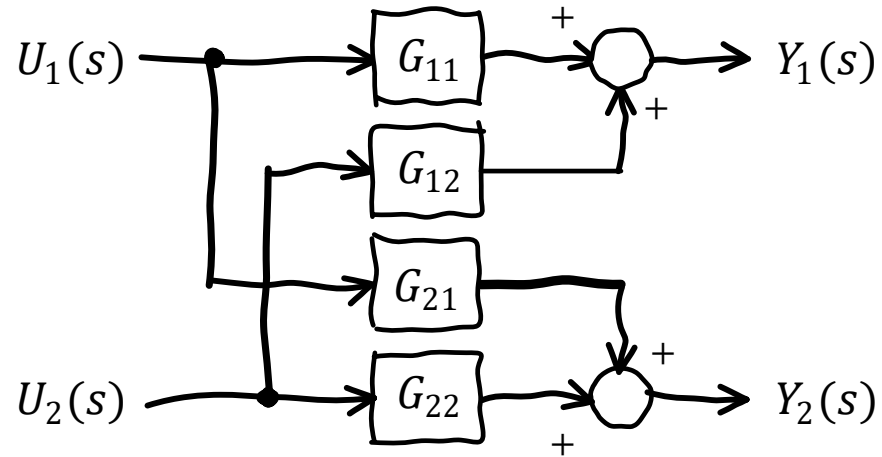
$$Y_k(s) = G_{k1}(s)U_1(s) + G_{k2}(s)U_2(s) + \dots + G_{km}(s)U_m(s), \quad k = 1, 2, \dots, r$$

- Vektorska notacija

$$\mathbf{Y}(s) = \mathbf{G}(s) \cdot \mathbf{U}(s)$$

$$\mathbf{G}(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \dots & G_{1m}(s) \\ G_{21}(s) & G_{22}(s) & \dots & G_{2m}(s) \\ \dots & \dots & \dots & \dots \\ G_{r1}(s) & G_{r2}(s) & \dots & G_{rm}(s) \end{bmatrix}, \mathbf{U}(s) = \begin{bmatrix} U_1(s) \\ U_2(s) \\ \dots \\ U_m(s) \end{bmatrix}, \mathbf{Y}(s) = \begin{bmatrix} Y_1(s) \\ Y_2(s) \\ \dots \\ Y_r(s) \end{bmatrix}$$

Primer funkcije prenosa sistema sa 2 ulaza i 2 izlaza



$$Y_1(s) = G_{11}(s)U_1(s) + G_{12}(s)U_2(s)$$

$$Y_2(s) = G_{21}(s)U_1(s) + G_{22}(s)U_2(s)$$

$$\mathbf{G}(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) \\ G_{21}(s) & G_{22}(s) \end{bmatrix}$$

Primetiti:

Ovo je linearan model.

Svaki izlaz se može dobiti kao suma ponašanja modela kada nezavisno deluju pojedinačni ulazi.

Znači, kada postavimo prvi ulaz, a drugi ulaz je nula dobijemo Y_{11} . Zatim, prvi ulaz je nula, a drugi postavimo i dobijemo y_{12} . Tada znamo da je $Y_1 = Y_{11} + Y_{12}$ kada deluju oba ulaza.

To važi za svaki izlaz, kao za modele sa više ulaza.

Dobijanje funkcije prenosa od linearnog matematičkog modela u prostoru stanja

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A} \cdot \hat{\mathbf{x}}(t) + \mathbf{B} \cdot \hat{\mathbf{u}}(t), \quad \hat{\mathbf{x}}(0) = \mathbf{0}$$

$$\hat{\mathbf{y}}(t) = \mathbf{C} \cdot \hat{\mathbf{x}}(t) + \mathbf{D} \cdot \hat{\mathbf{u}}(t)$$



Primeni se Laplasova transformacija
(početne vrednosti promenljivih stanja
su nula!)

$$s\mathbf{X}(s) - \mathbf{0} = \mathbf{A} \cdot \mathbf{X}(s) + \mathbf{B} \cdot U(s)$$

$$s\mathbf{I} \cdot \mathbf{X}(s) - \mathbf{A} \cdot \mathbf{X}(s) = \mathbf{B} \cdot U(s)$$

$$(s\mathbf{I} - \mathbf{A}) \cdot \mathbf{X}(s) = \mathbf{B} \cdot U(s)$$

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{B} \cdot U(s)$$

$$Y(s) = \mathbf{C} \cdot \mathbf{X}(s) + \mathbf{D} \cdot U(s)$$

$$Y(s) = (\mathbf{C} \cdot (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{B} + \mathbf{D})U(s)$$

$$\frac{Y(s)}{U(s)} = W(s) = \mathbf{C} \cdot (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{B} + \mathbf{D}$$

$$W(s) = \frac{P(s)}{Q(s)} = \frac{\mathbf{C} \cdot \text{adj}(s\mathbf{I} - \mathbf{A}) \cdot \mathbf{B} + \mathbf{D} \cdot \det(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}$$

Pažnja:

$\mathbf{I}$ je jedinična matrica.

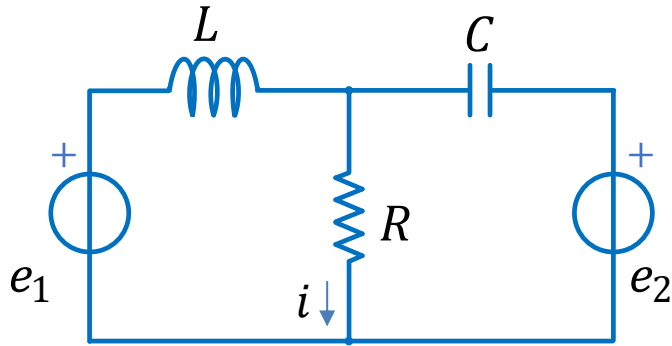
Množenje matrica nije komutativno!

Polovi sistema se dobijaju iz

$$\det(\lambda\mathbf{I} - \mathbf{A}) = 0$$

Rešenja su svojstvene
vrednosti $\lambda_i, i = 1, \dots, n$
matrice $\mathbf{A}$

Primer



Ulazi su naponi generatora,
a izlaz je struja kroz otpornik.

Napisati funkcije prenosa na osnovu linearnog matematičkog modela:

$$\begin{bmatrix} \frac{di_1}{dt} \\ \frac{du_C}{dt} \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

$$i = \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} i_1 \\ u_C \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$$

Izrada

Prepisujemo matrice:

$$A = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix}, \quad D = \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix}$$

Funkcija prenosa je

$$W(s) = C \cdot (sI - A)^{-1} \cdot B + D$$

...

$$A = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix}, \quad D = \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix}$$

$$(sI - A)^{-1} = \left(\begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \right)^{-1} = \begin{bmatrix} s & -\frac{1}{L} \\ \frac{1}{C} & s + \frac{1}{RC} \end{bmatrix}^{-1} = \frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \begin{bmatrix} s + \frac{1}{RC} & \frac{1}{L} \\ -\frac{1}{C} & s \end{bmatrix}$$

$$W(s) = C \cdot (sI - A)^{-1} \cdot B + D = \frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} s + \frac{1}{RC} & \frac{1}{L} \\ -\frac{1}{C} & s \end{bmatrix} \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix}$$

$$\begin{bmatrix} 0 & -\frac{1}{R} \end{bmatrix} \begin{bmatrix} s + \frac{1}{RC} & \frac{1}{L} \\ -\frac{1}{C} & s \end{bmatrix} = \begin{bmatrix} \frac{1}{RC} & -\frac{s}{R} \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{RC} & -\frac{s}{R} \end{bmatrix} \begin{bmatrix} \frac{1}{L} & -\frac{1}{L} \\ 0 & \frac{1}{RC} \end{bmatrix} = \begin{bmatrix} \frac{1}{RLC} & -\frac{1}{RLC} - \frac{s}{R^2C} \end{bmatrix}$$

$$W(s) = \frac{1}{s^2 + \frac{s}{RC} + \frac{1}{LC}} \begin{bmatrix} \frac{1}{RLC} & -\frac{1}{RLC} - \frac{s}{R^2C} \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{R} \end{bmatrix} = \begin{bmatrix} \frac{\frac{1}{RLC}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} & \frac{-\frac{1}{RLC} - \frac{s}{R^2C}}{s^2 + \frac{s}{RC} + \frac{1}{LC}} + \frac{1}{R} \end{bmatrix} = \begin{bmatrix} \frac{1}{RLCs^2 + Ls + R} & \frac{1}{R} - \frac{R + Ls}{R(RLCs^2 + Ls + R)} \end{bmatrix}$$

$$W(s) = \begin{bmatrix} \frac{1}{RLCs^2 + Ls + R} & \frac{LCs^2}{RLCs^2 + Ls + R} \end{bmatrix}$$

$$I(s) = W(s) \begin{bmatrix} E_1(s) \\ E_2(s) \end{bmatrix} = \frac{1}{RLCs^2 + Ls + R} E_1(s) + \frac{LCs^2}{RLCs^2 + Ls + R} E_2(s)$$

Šta smo dobili?

Zašto je dimenzija $W(s)$ 1x2?

Šta su ovde ulazi, izlazi?

Prednosti koncepta prostora stanja

- Koncept prostora stanja ima nekoliko prednosti u odnosu na klasični pristup (kompleksan domen), posebno ako se posmatra sa aspekta korišćenja digitalnih računara:
 - Određivanje rešenja sistema diferencijalnih jednačina prvog reda je brže na digitalnom računaru, nego rešavanje odgovarajuće diferencijalne jednačine višeg reda.
 - Uprošteno je matematičko opisivanje upotrebom vektorske notacije
 - Uključivanje početnih uslova sistema je jednostavno.
 - Model se može primeniti na vremenski promenljive, nelinearne, stohastičke i diskretne sisteme.

Matematički model u prostoru stanja

Izračunavanje kretanja promenljivih stanja i izlaza

Izračunavanje kretanja modela

- U vremenskom domenu

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A} \cdot \mathbf{x}(t) + \mathbf{B} \cdot \mathbf{u}(t), & \mathbf{x}(0) &= \mathbf{x}_0 \\ \mathbf{y}(t) &= \mathbf{C} \cdot \mathbf{x}(t) + \mathbf{D} \cdot \mathbf{u}(t)\end{aligned}$$

- Nakon primene Laplace-ove transformacije dobija se:
Početni uslovi ne moraju biti nula!

$$\begin{aligned}s\mathbf{X}(s) - \mathbf{x}(0) &= \mathbf{A} \cdot \mathbf{X}(s) + \mathbf{B} \cdot \mathbf{U}(s) \\ (s\mathbf{I} - \mathbf{A}) \cdot \mathbf{X}(s) &= \mathbf{x}(0) + \mathbf{B} \cdot \mathbf{U}(s) \\ \mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} \cdot \mathbf{U}(s)\end{aligned}$$

$$\begin{aligned}\mathbf{Y}(s) &= \mathbf{C} \cdot \mathbf{X}(s) + \mathbf{D} \cdot \mathbf{U}(s) \\ \mathbf{Y}(s) &= \mathbf{C} \cdot (s\mathbf{I} - \mathbf{A})^{-1} \cdot \mathbf{x}(0) + (\mathbf{C} \cdot (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}) \cdot \mathbf{U}(s)\end{aligned}$$

- Primena inverzne Laplasove transformacije na prethodni izraz daje kretanje izlaza modela ...

Izračunavanje kretanja modela (2)

- Računa se preko Fundamentalne matrice sistema: $\Phi(t) = \mathcal{L}^{-1}\{(s\mathbf{I} - \mathbf{A})^{-1}\}$
- Kretanje promenljivih stanja $\mathbf{x}(t)$

$$\mathbf{x}(t) = \underbrace{\Phi(t) \cdot \mathbf{x}(0)}_{\text{kretanje stanja sistema pod dejstvom početnih uslova}} + \underbrace{\int_0^t \Phi(t - \tau) \mathbf{B} u(\tau) d\tau}_{\text{kretanje stanja sistema pod dejstvom spoljne pobude}}$$

kretanje stanja sistema pod dejstvom spoljne pobude

kretanje stanja sistema pod dejstvom početnih uslova

- Kretanje izlaza $\mathbf{y}(t)$

$$\mathbf{y}(t) = \underbrace{\mathbf{C} \cdot \Phi(t) \cdot \mathbf{x}(0)}_{\text{kretanje izlaza sistema pod dejstvom početnih uslova}} + \underbrace{\mathbf{C} \cdot \int_0^t \Phi(t - \tau) \mathbf{B} u(\tau) d\tau}_{\text{kretanje izlaza sistema pod dejstvom spoljne pobude}} + \mathbf{D} \cdot u(t)$$

kretanje izlaza sistema pod dejstvom spoljne pobude

kretanje izlaza sistema pod dejstvom početnih uslova

Primer računanja odziva

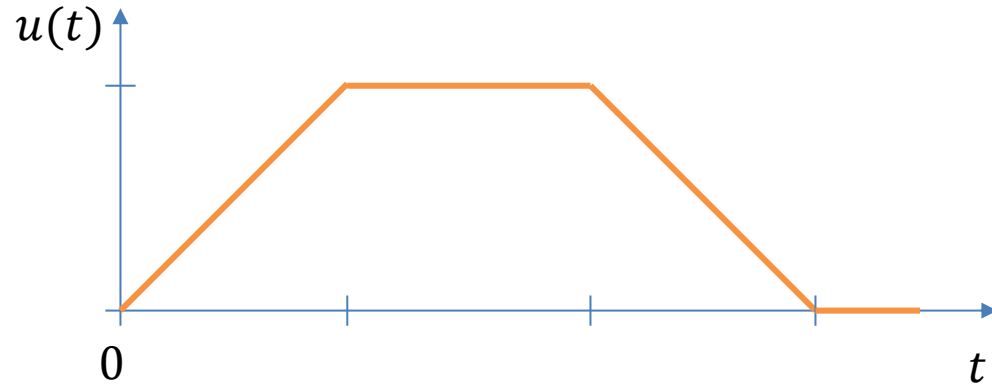
Dat je linearan vremenski kontinualan matematički model u prostoru stanja

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t), \quad \mathbf{x}(0) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$y(t) = [0 \quad 5] \mathbf{x}(t)$$

$$u(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & 1 \leq t < 2 \\ 3 - t, & 2 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$

Odrediti izlaz modela.



Matrice modela su:

$$\mathbf{A} = \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{C} = [0 \quad 5], \quad \mathbf{D} = 0$$

Primer računanja odziva – Fundamentalna matrica

Fundamentalna matrica:

$$\Phi(t) = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$$

$$(sI - A)^{-1} = \left(s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -3 & -2 \\ 1 & 0 \end{bmatrix} \right)^{-1} = \begin{bmatrix} s+3 & 2 \\ -1 & s \end{bmatrix}^{-1}$$

$$= \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s & -2 \\ 1 & s+3 \end{bmatrix} = \begin{bmatrix} \frac{s}{(s+1)(s+2)} & \frac{-2}{(s+1)(s+2)} \\ \frac{1}{(s+1)(s+2)} & \frac{s+3}{(s+1)(s+2)} \end{bmatrix} = \begin{bmatrix} \frac{-1}{s+1} + \frac{2}{s+2} & \frac{-2}{s+1} + \frac{2}{s+2} \\ \frac{1}{s+1} + \frac{-1}{s+2} & \frac{2}{s+1} + \frac{-1}{s+2} \end{bmatrix}$$

$$\Phi(t) = \begin{bmatrix} -e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & 2e^{-t} - e^{-2t} \end{bmatrix}$$

Primer računanja odziva – kretanje promenljivih stanja

Kretanje promenljivih stanja:

$$\mathbf{x}(t) = \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^t \mathbf{\Phi}(t - \tau)\mathbf{B}u(\tau)d\tau$$

$$\mathbf{x}(t) = \begin{cases} \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^t \mathbf{\Phi}(t - \tau)\mathbf{B}\tau d\tau, & 0 \leq t < 1 \\ \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^1 \mathbf{\Phi}(t - \tau)\mathbf{B}\tau d\tau + \int_1^t \mathbf{\Phi}(t - \tau)\mathbf{B}1 d\tau, & 1 \leq t < 2 \\ \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^1 \mathbf{\Phi}(t - \tau)\mathbf{B}\tau d\tau + \int_1^2 \mathbf{\Phi}(t - \tau)\mathbf{B}1 d\tau + \int_2^t \mathbf{\Phi}(t - \tau)\mathbf{B}(3 - \tau) d\tau, & 2 \leq t < 3 \\ \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^1 \mathbf{\Phi}(t - \tau)\mathbf{B}\tau d\tau + \int_1^2 \mathbf{\Phi}(t - \tau)\mathbf{B}1 d\tau + \int_2^3 \mathbf{\Phi}(t - \tau)\mathbf{B}(3 - \tau) d\tau, & t \geq 3 \end{cases}$$

Primer računanja odziva – kretanje izlaza

Kretanje izlaza:

$$y(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t)$$

$$y(t) = \begin{cases} \mathbf{C}\Phi(t)\mathbf{x}(0) + \int_0^t \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau, & 0 \leq t < 1 \\ \mathbf{C}\Phi(t)\mathbf{x}(0) + \int_0^1 \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau + \int_1^t \mathbf{C}\Phi(t-\tau)\mathbf{B}1 d\tau, & 1 \leq t < 2 \\ \mathbf{C}\Phi(t)\mathbf{x}(0) + \int_0^1 \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau + \int_1^2 \mathbf{C}\Phi(t-\tau)\mathbf{B}1 d\tau + \int_2^t \mathbf{C}\Phi(t-\tau)\mathbf{B}(3-\tau) d\tau, & 2 \leq t < 3 \\ \mathbf{C}\Phi(t)\mathbf{x}(0) + \int_0^1 \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau + \int_1^2 \mathbf{C}\Phi(t-\tau)\mathbf{B}1 d\tau + \int_2^3 \mathbf{C}\Phi(t-\tau)\mathbf{B}(3-\tau) d\tau, & t \geq 3 \end{cases}$$

Primer računanja odziva - rešenje

Rešenje:

$$y(t) = \begin{cases} -\frac{15}{4} + \frac{5}{2}t + 30e^{-t} - \frac{45}{4}e^{-2t}, & 0 \leq t < 1 \\ \frac{5}{2} + 5(6 - e)e^{-t} - \frac{5}{4}(9 - e^2)e^{-2t}, & 1 \leq t < 2 \\ \frac{45}{4} - \frac{5}{2}t + 5(6 - e + e^2)e^{-t} - \frac{5}{4}(9 - e^2 - e^4)e^{-2t}, & 2 \leq t < 3 \\ 5(6 - e - e^2 + e^3)e^{-t} - \frac{5}{4}(9 - e^2 - e^4 + e^6)e^{-2t}, & t \geq 3 \end{cases}$$

Primer računanja odziva – međurezultati ...

$$\mathbf{C}\Phi(t)\mathbf{x}(0) = \begin{bmatrix} 0 & 5 \end{bmatrix} \begin{bmatrix} -e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & 2e^{-t} - e^{-2t} \end{bmatrix} \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 & 5 \end{bmatrix} \begin{bmatrix} -5e^{-t} - 4e^{-2t} \\ 5e^{-t} - 2e^{-2t} \end{bmatrix} = 25e^{-t} - 10e^{-2t}$$

$$\mathbf{C}\Phi(t)\mathbf{B} = \begin{bmatrix} 0 & 5 \end{bmatrix} \begin{bmatrix} -e^{-t} + 2e^{-2t} & -2e^{-t} + 2e^{-2t} \\ e^{-t} - e^{-2t} & 2e^{-t} - e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 5e^{-t} - 5e^{-2t}$$

$$\int_0^t \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau = \int_0^t (5e^{-(t-\tau)} - 5e^{-2(t-\tau)})\tau d\tau = 5e^{-t} \int_0^t \tau e^{\tau} d\tau - 5e^{-2t} \int_0^t \tau e^{2\tau} d\tau$$

$$\int \tau e^{a\tau} d\tau = \frac{1}{a} \tau e^{a\tau} - \frac{1}{a} \int e^{a\tau} d\tau = \frac{1}{a} \tau e^{a\tau} - \frac{1}{a^2} e^{a\tau} = \frac{1}{a} \left(\tau - \frac{1}{a} \right) e^{a\tau}$$

$$\int_0^t \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau = 5e^{-t} (\tau e^{\tau} - e^{\tau}) \Big|_0^t - 5e^{-2t} \left(\frac{1}{2} \tau e^{2\tau} - \frac{1}{4} e^{2\tau} \right) \Big|_0^t$$

$$= 5e^{-t} (e^t(t-1) + 1) - 5e^{-2t} \left(\frac{1}{2} t e^{2t} - \frac{1}{4} e^{2t} + \frac{1}{4} \right) = 5(t-1+e^{-t}) - 5 \left(\frac{1}{2} t - \frac{1}{4} + \frac{1}{4} e^{-2t} \right) = \frac{5}{2} t - \frac{15}{4} + 5e^{-t} - \frac{5}{4} e^{-2t}$$

...

Primer računanja odziva – međurezultati.

...

$$\begin{aligned}\int_0^1 \mathbf{C}\Phi(t-\tau)\mathbf{B}\tau d\tau &= \int_0^1 (5e^{-(t-\tau)} - 5e^{-2(t-\tau)})\tau d\tau = 5e^{-t} \int_0^1 \tau e^{\tau} d\tau - 5e^{-2t} \int_0^1 \tau e^{2\tau} d\tau \\ &= 5e^{-t} (\tau e^{\tau} - e^{\tau}) \Big|_0^1 - 5e^{-2t} \left(\frac{1}{2} \tau e^{2\tau} - \frac{1}{4} e^{2\tau} \right) \Big|_0^1 = 5e^{-t} - 5e^{-2t} \left(\frac{1}{4} e^2 + \frac{1}{4} \right)\end{aligned}$$

$$\begin{aligned}\int_1^t \mathbf{C}\Phi(t-\tau)\mathbf{B}1 d\tau &= \int_1^t (5e^{-(t-\tau)} - 5e^{-2(t-\tau)}) d\tau = 5e^{-t} \int_1^t e^{\tau} d\tau - 5e^{-2t} \int_1^t e^{2\tau} d\tau \\ &= 5e^{-t} (e^{\tau}) \Big|_1^t - \frac{5}{2} e^{-2t} (e^{2\tau}) \Big|_1^t = 5e^{-t}(e^t - e) - \frac{5}{2} e^{-2t}(e^{2t} - e^2) = \frac{5}{2} - 5e^{-(t-1)} + \frac{5}{2} e^{-2(t-1)}\end{aligned}$$

$$\int_1^2 \Phi(t-\tau)\mathbf{B}1 d\tau = 5e^{-t} (e^{\tau}) \Big|_1^2 - \frac{5}{2} e^{-2t} (e^{2\tau}) \Big|_1^2 = 5e^{-t}(e^2 - e) - \frac{5}{2} e^{-2t}(e^4 - e^2)$$

$$\int_2^t \Phi(t-\tau)\mathbf{B}(3-\tau) d\tau = \frac{15}{4} e^{4-2t} - 10e^{2-t} - \frac{5}{2} t + \frac{45}{4}$$

$$\int_2^3 \Phi(t-\tau)\mathbf{B}(3-\tau) d\tau = \frac{5}{4} e^{-2(t-1)}(3e^2 - e^4 - 8e^t + 4e^{t+1})$$